

Healthcare – Persistency of a Drug (Data Science Specialization)

Week 8: Data Understanding & Data Issues

Team Member's Details

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Date	23 October 2025

Problem Description

The pharmaceutical industry continues to face significant challenges in ensuring that patients remain adherent to their prescribed therapies. Non-adherence or premature discontinuation of medication leads to poor treatment outcomes, increased costs, and higher risks of hospital readmission.

To address this issue, ABC Pharma Company has adopted a data-driven approach to analyse the persistency behaviour of patients taking a specific drug.

The main objective of this project is to analyse and model patient-level data in order to identify the factors that influence whether a patient continues or discontinues their medication. The target variable, `Persistency_Flag`, represents this behaviour:

- 1 = Persistent – the patient continued therapy for the intended duration
- 0 = Non-persistent – the patient stopped taking the drug early

By analysing the dataset, the goal is to provide actionable insights that can help healthcare providers, insurers, and pharmaceutical companies design targeted interventions that improve patient retention and health outcomes.

Business Understanding

From a business standpoint, patient persistency directly impacts both medical outcomes and organizational profitability.

For pharmaceutical firms, higher persistency translates to improved therapeutic success and predictable revenue.

For healthcare systems, it reduces unnecessary complications and long-term treatment costs.

Through data analysis and predictive modelling, this project aims to:

- Identify key demographic, clinical, and behavioral factors affecting persistency.
- Predict which patients are at high risk of non-adherence early in treatment.
- Enable data-driven decision-making for patient follow-up and engagement.
- Improve resource allocation by focusing on populations that need additional support.

Data Understanding

Dataset Overview

The dataset – Healthcare_dataset.xlsx – contains 3424 rows and 27 columns, combining information about patient demographics, medical characteristics, prescriber details, risk segments, and adherence behavior.

Category	Example Features	Description
Demographics	Gender, Age, Ethnicity, Race, Region	Patient-level descriptive attributes
Physician Data	Ntm_Speciality, Ntm_Specialist_Flag	Type and specialization of prescribing physician
Risk and Clinical	Risk_Segment_Prior_Ntm, Risk_Segment_During_Rx, T_Score_Change	Indicators of medical risk and therapy response
Adherence Metrics	Adherent_Flag, Change_Risk_Segment	Tracks changes and medication behavior
Target Variable	Persistency_Flag	1 = Persistent, 0 = Non-persistent

Initial Inspection

Exploration of the dataset in Jupyter Notebook revealed:

- No missing attributes or completely empty rows.
- Categorical values encoded inconsistently (e.g., “Y/N”, “Yes/No”, “Unknown”).
- Slight skewness observed in continuous variables such as Age, Risk_Score, and T_Score_Change.
- A small proportion of outliers detected in numerical variables, particularly in Persistency_Duration and Risk_Score.
- The dataset is overall clean and suitable for analysis but requires transformation to prepare it for modeling.

Problems in the Data

Issue Type	Description	Example Columns
Missing Values	No explicit NAs, but several “Unknown” category values act as implicit missing data.	Race, Risk_Segment_Prior_Ntm
Outliers	Unusually high durations or risk scores beyond typical distribution range.	Persistency_Duration, Risk_Score
Skewness	Right-skewed distributions with long tails.	Age, Risk_Score
Inconsistent Encoding	Different categorical encodings for similar meaning (e.g., Y/N vs Yes/No).	Adherent_Flag, Change_Risk_Segment
Unbalanced Classes	Non-persistent patients appear slightly more frequent than persistent ones.	Persistency_Flag

A closer look showed that while missing values were minimal, the presence of “Unknown” categories could distort summary statistics and model accuracy if not treated properly.

Approaches to Handle Data Problems and Rationale

Data Problem	Proposed Approach	Why This Approach is Used
Missing or 'Unknown' Values	Replace "Unknown" with NaN, then impute using mode (for categorical) or median (for numerical) values.	Preserves dataset size while reducing bias caused by ambiguous categories.
Outliers	Apply IQR (Interquartile Range) capping method to limit extreme values or remove if biologically implausible.	Prevents extreme points from dominating model coefficients.
Skewness	Use log transformation or Box-Cox normalization on continuous features.	Improves normality assumption for algorithms such as logistic regression.
Categorical Encoding Issues	Standardize all binary fields to numeric (1 = Yes, 0 = No); apply One-Hot Encoding to multi-class categorical features.	Ensures algorithm compatibility and interpretability.
Class Imbalance	Apply resampling (SMOTE / undersampling) during model training.	Balances target classes for fair prediction performance.
Feature Scaling	Normalize numerical values using StandardScaler or MinMaxScaler.	Aligns variable ranges for distance-based algorithms.

Exploratory Data Insights (Preliminary)

- **Gender Distribution:** Females form a larger portion of both persistent and non-persistent groups.
- **Age Pattern:** Older patients (> 65) show slightly higher persistency but also more variability.
- **Ethnicity and Race:** Non-Hispanic Caucasian patients dominate the dataset; this must be considered to avoid bias.
- **Provider Type:** Patients under general practitioners and rheumatologists showed the highest counts for both persistency categories.
- **Risk Profile:** Low-risk patients were more likely to discontinue therapy early than high-risk patients.

These insights were supported through visualization in Python using **Matplotlib** and **Seaborn** during early EDA.

Data Summary

Metric	Value
Total Observations	3 424
Total Features	27
File Format	.xlsx
Size	899 KB
Missing Values	0 explicit NAs / $\approx 5\%$ implicit 'Unknowns'
Outliers	3–5 % of rows in numeric features
Skewness	Mild-to-moderate positive skew in Age & Risk_Score

Next Steps

1. Implement data cleansing using the chosen methods above.
2. Transform categorical data into numeric format.
3. Conduct feature selection based on correlation and variance.
4. Perform full Exploratory Data Analysis (EDA) with visual insights and statistical tests.
5. Prepare dataset for model training using classification algorithms such as Logistic Regression, Decision Tree, and Random Forest.

GitHub Repository Link

<https://github.com/magdalena-ivanova-ds/Data-Glacier-Virtual-Internship/tree/main/week8>